9.1 Sample Spaces and Probability

Name _____ Period ____ Date _____

Algebra I 9.1 (30-minute plan) · Student sheet

I can…

- list the outcomes in a sample space and count them.
- find theoretical probabilities and the probability of a complement.
- find geometric and experimental probabilities.

Vocabulary · fill in the missing word

Probability experiment — an action or trial with _____ results

Outcome — one possible result of an _____

Event — a _____ of one or more outcomes

Sample space — the set of all _____ outcomes

Probability — a number from 0 (_____) to 1 (certain) that measures how likely an event is

Theoretical probability — (number of _____ outcomes) ÷ (total number of outcomes), when outcomes are equally likely

Complement of A — all _____ not in A; P(not A) = 1 − P(A)

Geometric probability — a probability found from _____ or areas

Experimental probability — (number of _____) ÷ (number of trials)

Part 1 · Notes and examples (fill in each note, then work the example)

Reading a probability: Probability 0 means _____; probability 1 means _____.

Example 1

You flip a coin and spin a spinner with three equal sections: red, blue, and green. List the sample space. How many outcomes are there?

Theoretical: P(A) = _____ ÷ _____ **Complement:** P(not A) = _____

Example 2

You guess on three true-false questions. Find the probability that you get

a. all three correct. b. exactly two correct.

Geometric: P = (favorable _____ or area) ÷ (total length or _____)

Experimental: P = _____ ÷ _____

Example 3

A dart lands at a random point on a square board 20 inches on a side. A circle with a radius of 5 inches is in the center. What is the probability that the dart lands in the circle?

Part 2 · Partner practice

Partner practice

A spinner has 8 equal sections numbered 1 to 8. Find P(a number greater than 5) and P(a number not greater than 5).